From (1) and (2),

x and $b + c$ both are equal.

Therefore, $x = b + c$.

In the same way, we can prove the result for other exterior angles.

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Now we are going to learn one more result on the exterior angle of a triangle for which we do the following activity.

Activity

Imagine a person standing in one of the vertices (corners) and walking along the boundary of the triangle until he reaches the starting point. At each of the vertex of the triangle he would turn an angle equal to the exterior angle at that vertex. Hence after the complete journey around the triangle he would have turned through an angle equal to one complete revolution, that is 360°.

We prove this result as below.

Since the angles on a straight line is 180° ,

We have $a + x = 180°$ [linear pair of angles are supplementary]

$x = 180° - a$

Similarly, $y = 180° - b$

and $z = 180° - c$

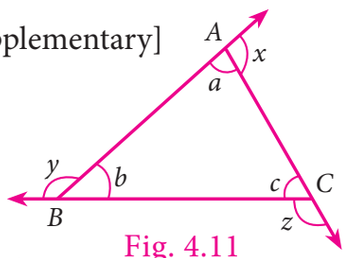


Fig. 4.11

Therefore, $x + y + z = (180° - a) + (180° - b) + (180° - c)$

$= 540 - (a + b + c)$

$= 540° - 180°$ [sum of all three angles of a triangle is 180°]

$= 360°$

Hence, the sum of all exterior angles of a triangle is 360°.

Therefore, the properties of exterior angle gives us the following two results.

(i) An exterior angle of a triangle is equal to the sum of its interior opposite angles.

(ii) The sum of exterior angles of a triangle is 360° .

Example 4.5 In ΔPQR, find the exterior angle, $\angle SRQ$.

Solution

Let, $\angle SRQ = x$

Exterior angle = sum of two interior opposite angles

$x = 38° + 44° = 82°$

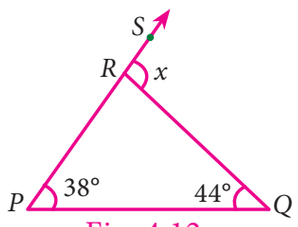


Fig. 4.12

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